

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250271791

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

WATANABE; YUKIMASA et al.

MAGNET ROLLER, DEVELOPING ROLLER, AND IMAGE FORMING APPARATUS

Abstract

A magnet roller includes a shaft, a plurality of magnets, and an adhesive. The plurality of magnets is arranged in a circumferential direction of a shaft center of the shaft on an outer surface of the shaft. The adhesive bonds the shaft and the plurality of magnets to each other. At least one of the magnets has a recess portion recessed with respect to a surface of the shaft, the surface facing the outer surface.

Inventors: WATANABE; YUKIMASA (OSAKA, JP), EZURE; KAZUAKI (OSAKA, JP), SATO; HARUHIRO (OSAKA, JP)

Applicant: KYOCERA DOCUMENT SOLUTIONS INC. (OSAKA, JP)

Family ID: 1000008491241

Appl. No.: 19/059280

Filed: February 21, 2025

Foreign Application Priority Data

JP	2024-025838	Feb. 22, 2024
----	-------------	---------------

Publication Classification

Int. Cl.: G03G15/09 (20060101); H01F7/02 (20060101)

U.S. Cl.:

CPC G03G15/0928 (20130101); H01F7/0221 (20130101); G03G2215/0636 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION

[0001] This application claims the benefit of Japanese Priority Patent Application JP 2024-025838 filed Feb. 22, 2024, under 35 U.S.C. 119, the entire contents of which are incorporated herein by reference.

FIELD OF THE DISCLOSURE

[0002] The present disclosure relates a magnet roller, a developing roller, and an image forming apparatus.

BACKGROUND OF THE DISCLOSURE

[0003] Conventionally, in a developing part of an image forming apparatus, a developer is supplied to a photosensitive drum and toner images are formed on the photosensitive drum. Specifically, the developing part supplies the developer to the photosensitive drum through a developing roller. Japanese Patent Application Laid-open No. 2022-20305 has described a magnet roller that is used in the developing roller.

[0004] The magnet roller of the developing roller described in Japanese Patent Application Laid-open No. 2022-20305 includes a plurality of magnetic poles. The developing roller further includes a sleeve that covers the outer circumference of the magnet roller to be rotatable. The plurality of magnetic poles is formed of a plurality of different magnets. The magnet roller is assembled by bonding the plurality of magnets to the shaft.

SUMMARY OF THE DISCLOSURE

[0005] In accordance with one aspect of the present disclosure, a magnet roller includes a shaft, a plurality of magnets, and an adhesive. The plurality of magnets is arranged in a circumferential direction of a shaft center of the shaft on an outer surface of the shaft. The adhesive bonds the shaft and the plurality of magnets to each other. At least one of the plurality of magnets has a recess portion that is recessed with respect to a surface facing the outer surface of the shaft.

[0006] In accordance with another aspect of the present disclosure, a developing roller includes such a magnet roller and a tubular sleeve. The sleeve houses the magnet roller and is rotatably supported around the circumferential direction.

[0007] In accordance with still another aspect of the present disclosure, an image forming apparatus includes such a developing roller.

[0008] In accordance with the present disclosure, a technology capable of improving the productivity of the developing roller can be provided.

[0009] These and other objects, features and advantages of the present disclosure will become more apparent in light of the following detailed description of best mode embodiments thereof, as illustrated in the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a view showing an internal configuration of an image forming apparatus with a magnet roller according to a first embodiment of the present disclosure.

[0011] FIG. 2 is a view schematically showing a configuration of a developing part with the magnet roller.

[0012] FIG. 3 is a perspective view showing a configuration of the magnet roller.

[0013] FIG. 4 is a perspective view showing a configuration of a magnet of the magnet roller.

[0014] FIG. 5A is a view showing an assembling procedure of the magnet roller.

[0015] FIG. 5B is a view showing the assembling procedure of the magnet roller.

[0016] FIG. 5C is a view showing the assembling procedure of the magnet roller.

[0017] FIG. 6 is a plan view showing a configuration of a magnet roller according to a second

embodiment of the present disclosure.

[0018] FIG. 7A is a view showing a test method for checking bond strength between the magnet and the shaft.

[0019] FIG. 7B is a diagram showing an example of measurement results of magnetic properties of the magnet roller.

[0020] FIG. 8A is a diagram showing an example of measurement results of the bond strength and the magnetic properties of the magnet.

[0021] FIG. 8B is a diagram showing an example of measurement results of the bond strength and the magnetic properties of the magnet.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

[0022] Hereinafter, a magnet roller **1**, **1d** according to first and second embodiments of the present disclosure will be described with reference to the drawings. Note that in the figures, the same or corresponding portions will be denoted by the same reference signs and the descriptions will not be repeated. Moreover, in the following description, terms meaning positions and directions, e.g., “upper,” “lower,” “lateral,” and “vertical” may be used. These terms are used for the sake of convenience to make the embodiment easily understood, and they are not limited to positions or directions when it is actually carried out.

First Embodiment

[0023] An image forming apparatus **100** with the magnet roller **1** according to the first embodiment of the present disclosure will be described with reference to FIG. **1**. FIG. **1** is a view showing an internal configuration of the image forming apparatus **100** with the magnet roller **1** according to the first embodiment of the present disclosure.

[0024] According to FIG. **1**, the image forming apparatus **100** includes paper feed devices **61**, conveyance paths **62**, conveyance rollers **63**, an image forming part **5**, a fixation part **64**, and a delivery tray **65**. The paper feed devices **61** feeds sheets **S**. The sheets **S** fed from the paper feed devices **61** are conveyed on the conveyance paths **62**. The conveyance rollers **63** convey the sheets **S** on the conveyance paths **62**. The image forming part **5** is provided in the middle way of the conveyance paths **62** and forms a toner image on each of the sheets **S**. The fixation part **64** fixes the toner image formed on the sheet **S**. The delivery tray **65** delivers the sheet **S** on which the image is fixed.

[0025] The image forming part **5** includes photosensitive drums **8**, exposure parts (not shown), developing parts **4**, and transferring rollers **60**. Toner images corresponding to image data are formed on surface layers of the photosensitive drums **8** by the exposure parts and the developing parts **4**. Note that the image data is image data scanned by, for example, a script scanning part **67** or image data received by an external communication unit **68**. A control unit **69** controls the external communication unit **68**.

[0026] The transferring rollers **60** are brought into press-contact with the photosensitive drums **8** and form transferring nip regions **N1** between the transferring rollers **60** and the photosensitive drums **8**. A transferring voltage is applied on the surfaces of the transferring rollers **60**. When the toner images on the surfaces of the photosensitive drums **8** pass through the transferring nip regions **N1** together with the sheet **S**, the toner images are transferred to the sheet **S** and images are formed on the sheet **S** by electrostatic attraction of the transferring rollers **60**. The sheet **S** on which the images are formed is conveyed to the fixation part **64**.

[0027] The fixation part **64** includes a fixation roller **64b** in which a heater is built and a press roller **64c**. The press roller **64c** is brought into press-contact with the fixation roller **64b** and forms a fixation nip region **N2** between the press roller **64c** and the fixation roller **64b**. When the sheet **S** on which the images are formed passes through the fixation nip region **N2**, the images are fixed on the sheet **S** by press and heating by the rollers **64b** and **64c**. The sheet **S** on which the images are fixed is conveyed to the delivery tray **65**.

[0028] Next, a configuration of the developing part **4** will be described with reference to FIG. **2**.

FIG. 2 is a view schematically showing a configuration of the developing part 4 with the magnet roller 1 according to the present disclosure.

[0029] In FIG. 2, the developing part 4 includes a developer storage part 6, a developing roller 3, and a regulating member 7. The developer storage part 6 stores a developer A. The developing roller 3 adsorbs the developer A from the developer storage part 6. The regulating member 7 rubs off the amount of the developer A absorbed by the developing roller 3 and regulates it to a certain amount.

[0030] The developing roller 3 supplies the absorbed and regulated developer A to the photosensitive drums 8. The developing roller 3 includes the magnet roller 1 and a sleeve 2. The magnet roller 1 has a plurality of magnetic poles. The sleeve 2 covers the outer periphery of the magnet roller 1 to be rotatable. The magnet roller 1 includes a columnar shaft 30 and a plurality of magnets 10. The plurality of magnets 10 is arranged in a circumferential direction D2 of the shaft 30 (an example of a circumferential direction of a shaft center).

[0031] The magnets 10 are provided to face an outer surface 30e of the shaft 30 and extends in an axial direction D1 of the shaft 30. The plurality of magnets 10 has inner surfaces that face the outer surface 30e of the shaft 30. The inner surfaces of the plurality of magnets 10 and the outer surface 30e of the shaft 30 are bonded to each other with an adhesive (not shown).

[0032] In this embodiment, the magnets 10 include a first magnet 11, a second magnet 21, a third magnet 31, a fourth magnet 41, and a fifth magnet 51. In this embodiment, the first magnet 11, the third magnet 31, and the fifth magnet 51 have their N poles facing radially outward, while the second magnet 21 and the fourth magnet 41 have their S poles facing radially outward.

[0033] Each of the magnetic poles included in the magnetic roller 1 has its own role and is required to make the magnetic force pattern on the surface of the sleeve 2 into a desired magnetic force waveform.

[0034] Specifically, each magnet such as the first magnet 11 corresponds to one of a pumping pole, a conveyance pole, a regulating pole, a developing pole, and a stripping pole. The pumping pole, the conveyance pole, the regulating pole, the developing pole, and the stripping pole are arranged along an arrow D4 direction (direction of rotation of the sleeve 2) shown in FIG. 2 in the stated order.

[0035] The developer A is constituted by toner and carrier (magnetic particles). The carrier is magnetically adsorbed on the magnetic roller 1 while carrying the toner. The developer A is charged by a charging apparatus (not shown), and is magnetically adsorbed on the pumping pole and adheres to the sleeve 2. The conveyance pole conveys the developer A that has adhered to the sleeve 2 toward the regulating pole.

[0036] The conveyed developer A is rubbed off by the regulating member 7, and the conveyance amount of the developer A becomes uniform. The developer A is conveyed to the developing pole after passing through the regulating pole. At the developing pole, the toner of the developer A is magnetically adsorbed and adheres to the photoconductor drum 8 rotating in an arrow D5 direction shown in FIG. 2. An electrostatic latent image corresponding to the image data has been formed on the surface layers of the photosensitive drum 8 by charging with the charging apparatus and exposure to light such as irradiation with laser light. The toner is adsorbed at a position of the surface layer of the photosensitive drum 8, where the electrostatic latent image has been formed. Therefore, the toner images corresponding to the image data is developed on the surface layer of the photosensitive drum 8.

[0037] Then, the developer A remaining on the sleeve 2 is conveyed to the stripping pole. The stripping pole is configured so that the magnetic field becomes weaker, and the developer A conveyed to the stripping pole is peeled off from the sleeve 2.

[0038] Next, a configuration of the magnet roller 1 will be further described with reference to FIG. 3. FIG. 3 is a perspective view showing the configuration of the magnet roller 1.

[0039] In FIG. 3, the shaft 30 has a cutout portion 30n on a part of the outer surface 30e at the end

portion in the axial direction D1. The cutout portion **30n** serves as a reference for positioning the shaft **30** in the circumferential direction D2.

[0040] In order to form a desired magnetic force pattern on the surface of the sleeve **2** (see FIG. 2) over the circumferential direction D2 of the shaft **30**, each magnet needs to be accurately positioned in the circumferential direction D2 and bonded to the shaft **30**. In this embodiment, each of the first magnet **11**, the second magnet **21**, the third magnet **31**, the fourth magnet **41**, and the fifth magnet **51** needs to be accurately positioned on the basis of the reference cutout portion **30n** of the shaft **30**.

[0041] Specifically, the first magnet **11** is first bonded to the shaft **30** at a desired position in the circumferential direction D2 on the basis of the reference cutout portion **30n**. Then, the second magnet **21**, the third magnet **31**, the fourth magnet **41**, and the fifth magnet **51** are sequentially bonded to the shaft **30** on the basis of the positions of side surfaces **15** of the first magnet **11** as a reference (to be described later in detail). The first magnet **11**, the second magnet **21**, the third magnet **31**, the fourth magnet **41**, and the fifth magnet **51** are bonded in a state in which they are arranged at desired positions in the circumferential direction D2 on the basis of the position of the cutout portion **30n** as a reference. A desired magnetic pole pattern is thus formed on the surface of the sleeve **2** that covers the magnet roller **1**.

[0042] Next, a configuration of the first magnet **11** of the magnet roller **1** according to the present disclosure will be described with reference to FIG. 4. FIG. 4 is a perspective view showing the configuration of the first magnet **11** of the magnets **10** of the magnet roller **1**.

[0043] As shown in FIG. 4, the first magnet **11** has an inner surface **12** (an example of a surface), an outer surface **16**, and two side surfaces **15**. A cross-section of the inner surface **12**, which faces the outer surface **30e** of the shaft **30**, has a circular arc-shape. The outer surface **16** is located outside the inner surface **12**. The outer surface **16** has a curve line-shaped cross-section. The two side surfaces **15** connect the inner surface **12** and the outer surface **16** to each other in a radial direction D3 of the shaft **30**.

[0044] Two grooves **13** (an example of a plurality of recess portions) extending in the axial direction D1 are provided in the inner surface **12**. The two grooves **13** are provided at positions spaced apart from each other in the circumferential direction D2 on the inner surface **12**. The inner surface **12** is divided into an inner surface **12i** at the inside in the circumferential direction D2 and an inner surface **12e** at the outside by the two grooves **13**.

[0045] Next, an assembling procedure of the magnet roller **1** described above and actions thereof will be described with reference to FIGS. 5A, 5B, and 5C. FIGS. 5A, 5B, and 5C are views showing the assembling procedure of the magnet roller **1**.

[0046] As described above, the first magnet **11** and the shaft **30** are first bonded when bonding the magnet **10** to the shaft **30**. Then, the second magnet **21**, the third magnet **31**, the fourth magnet **41**, the fifth magnet **51**, and the shaft **30** are sequentially bonded.

[0047] As shown in FIG. 5A, an adhesive G is applied to the inner surface **12i** inside the first magnet **11** when bonding the first magnet **11** to the shaft **30**. Then, the shaft **30** is pushed against the inner surface **12** of the first magnet **11** as shown by the virtual lines in the same figure. When bonding the first magnet **11** to the shaft **30**, the position of the first magnet **11** in the circumferential direction D2 with respect to the shaft **30** is accurately positioned when the outer surface **30e** of the cutout portion **30n** is pushed against the inner surface **12** of the first magnet **11**.

[0048] Note that a designed value of the amount of application of the adhesive G is favorably determined to obtain a desired bonding thickness so as to maximize the bond strength. The designed value of the amount of application of the adhesive G is favorably set with a constant width in accordance with requirements (task reliability, costs, etc.) desired for production.

[0049] The second and subsequent magnets **21** are sequentially bonded by using the side surfaces **15** of the first magnet **11**, whose bonding has been completed, as a reference surface for positioning. Therefore, it is favorable that the adhesive G solidifies and the bonding between the first magnet **11** and the shaft **30** is completed before the second magnet **21** is bonded.

[0050] FIG. 5B shows a state in which the bonding between the first magnet **11** and the shaft **30** has been completed. As shown in an enlarged view of FIG. 5B, the excess adhesive G from the applied adhesive G flows into the grooves **13** and does not leak out of the grooves **13**. Accordingly, the adhesive G does not leak to the side surfaces **15** of the first magnet **11**, as is conventionally the case. If the width and depth of the grooves **13** are set appropriately, the magnet roller **1** can be configured so that the adhesive G does not leak to the side surfaces **15** of the magnet **10** within a width that is the designed value of the amount of application of the adhesive G.

[0051] In addition, as shown in FIG. 5C, when the second magnet **21** is bonded, the side surfaces **15** of the first magnet **11**, which has been already positioned, serve as a reference for positioning the second magnet **21** in the circumferential direction D2. Therefore, the second magnet **21** is accurately positioned and bonded to the shaft **30**. As a result, the occurrence of gaps and floating as seen in a case of the conventional magnet can be suppressed. Since the second magnet **21** is accurately positioned, the third and subsequent magnets **31** can also be accurately positioned and bonded to the shaft **30**.

[0052] As a result, the magnet roller **1** can suppress a reduction in bond strength because the bonding area between the magnets **10** and the shaft **30** does not decrease, and it can prevent the magnets **10** from being peeled off from the shaft **30** at the time of assembling the developing roller **3**, for example. That is, the magnet roller **1** can improve the productivity of the developing roller **3**.

[0053] In a case where the plurality of magnets is conventionally bonded to the shaft, if the amount of application of the adhesive to the magnets is large, the adhesive cures in a state of leaking to the side surfaces of the magnets. In this case, there can be floating between the shaft and the other magnets. This may cause a defect in assembling the developing roller, which may reduce the productivity of the developing roller. On the other hand, the above-mentioned magnet roller **1** can improve the productivity of the developing roller.

[0054] As in this embodiment, the two grooves **13** are provided at positions spaced apart from each other in the circumferential direction D2 on the inner surface **12** of the first magnet **11** at a distance from each other in the circumferential direction D2, thereby forming the inner surface **12i** at the inside on the magnet **10**. By forming the inner surface **12i** at the inside, the leakage of the adhesive G is suppressed from both sides of the inner surface **12** in the circumferential direction D2, and thus the leakage of the adhesive G to the side surfaces **15** of the first magnet **11** is surely suppressed.

[0055] Although the shaft **30** according to this embodiment is configured to have a columnar shape, it may also be configured to have a polygonal cylindrical shape, such as a square cylindrical shape. In that case, the inner surfaces of the magnets **10**, which face the outer surface **30e** of the shaft **30**, are favorably configured to have a shape corresponding to the shape of the outer surface **30e** of the shaft **30**.

[0056] Although the magnet roller **1** according to this embodiment is configured so that the grooves **13** are provided only in the first magnet **11** of the magnets **10**, the magnet with the grooves **13** is not limited to the first magnet **11**. The magnet roller **1** may be configured so that the grooves **13** are provided in any magnet of the second magnet **21**, the second magnet **21**, the third magnet **31**, the fourth magnet **41**, and the fifth magnet **51**.

[0057] Although the grooves **13** are formed on the inner surface **12** of the magnet **10** in the magnet roller **1** according to this embodiment, the grooves **13** may be formed on the outer surface **30e** of the shaft **30** or may be formed on both the inner surface of the magnet **10** and the outer surface **30e** of the shaft **30**.

[0058] Also in a case where the grooves **13** are formed on the outer surface of the shaft **30**, the excess adhesive G of the applied adhesive G flows in the grooves **13** as described above, and does not leak out of the grooves **13**. Therefore, the magnet roller **1** can provide actions and effects similar to the above-mentioned actions and effects.

Second Embodiment

[0059] Next, a configuration of the magnet roller **1d** according to the second embodiment will be described with reference to FIG. 6. FIG. 6 is a plan view showing a configuration of the magnet roller **1d** according to the second embodiment of the present disclosure.

[0060] As shown in FIG. 6, a first magnet **11d** of the magnet roller **1d** further defines the range of a groove width **13w** and a groove depth **13d** with respect to the first magnet **11** of the magnet roller **1** according to the first embodiment.

[0061] Specifically, a ratio of the sum of the widths of the plurality of grooves **13** in the circumferential direction D2 (hereinafter, referred to as “groove width ratio”) to a width **12w** of the inner surface **12** of the first magnet **11d** in the circumferential direction D2 of the shaft **30** is 12% or more and 25% or less.

[0062] Moreover, a ratio of a depth (hereinafter, referred to as “groove depth ratio”) of each of the plurality of grooves **13** to a height **15h** (an example of a dimension) of the first magnet **11d** in the radial direction D3 of the shaft **30** is 7% or more and 14% or less.

[0063] Hereinafter, the actions by the configuration of the above-mentioned magnet roller **1d** will be described, referring to test results related to the bond strength of the magnet roller **1d** and the magnetic properties of the magnet roller **1d**.

[0064] First of all, respective test methods will be described with reference to FIGS. 7A and 7B. FIG. 7A is a view showing the test method for checking the bond strength of the magnet roller **1d**. FIG. 7B is a diagram showing an example of measurement results of the magnetic properties of the magnet roller **1d**.

[0065] The bond strength is measured on the basis of, for example, “Tensile Bond Strength Test Methods for Adhesives” (JIS K 6849-1994). Specifically, as shown in FIG. 7A, the bond strength tests are performed by supporting the first magnet **11d** bonded to the shaft **30** with a jig J and using a push-pull gage P (“PS-10 Kg” manufactured by IMADA-SS Corporation). The push-pull gage P pulls the shaft **30** away from the first magnet **11d**, and the maximum load required for separating the shaft **30** from the first magnet **11d** is measured as “bond strength.”

[0066] Although the illustration is omitted, a measurement element using a Hall element is disposed on a magnet **10d** bonded to the shaft **30**, and the magnetic properties of the bonded magnet **10d** are detected by rotating the shaft **30**. Then, the relationship between the rotation angle of the shaft **30** and the magnetic flux density detected by the Hall element is obtained as shown in FIG. 7B. Specifically, an “automatic magnetic field distribution measurement apparatus: 6800ROLL2” manufactured by Nihon Denji Sokki co.,ltd. is used as a measurement apparatus. The measurement element is arranged at a position 10 mm from the center of the shaft **30** in the radial direction D3, and the rotational speed of the shaft **30** is set to be about 6 seconds per rotation. Under the same conditions as the above-mentioned conditions, the measurement is also performed in a case where the measurement elements are arranged at 9 positions in the axial direction D1 of the magnet **10**, which divide the length of the magnet **10** into 10 equal parts.

[0067] In the measurement waveform obtained as shown in FIG. 7B, the maximum value of the magnetic flux density waveform shown in the figure is a “maximum magnetic force.” Then, the rotation angle at the center of the two rotation angles $\theta 3$ and $\theta 4$, where the magnetic flux density is 80% of the maximum magnetic force, is determined as an “magnetic center angle. That is, these “maximum magnetic force” and “magnetic center angle” are checked as the magnetic properties.

[0068] Next, referring again to FIG. 6, the details of the first magnet **11d**, the shaft **30**, and the adhesive G used in the test will be described. The first magnet **11d** has a fan-shape with an opening angle $\theta 1$ of 90° shown in the cross-section. The height **15h** of the first magnet **11d** is 5.7 mm, the axial length is 300 mm, the outer shape of the circular arc forming the inner surface **12** is $\Phi 6$ mm, and the outer shape of the circular arc forming the outer surface **16** is $\Phi 17.4$ mm. The grooves **13** are arranged at positions where the angle $\theta 2$ in the circumferential direction D2 is 25° with respect to a center line C of the first magnet **11d** in the circumferential direction D2, respectively. The groove depth **13d** of the grooves **13** is 0.3 mm and the groove width **13w** is 0.6 mm. The material

of the first magnet **11d** contains polychloroethylene as a main component and 85 mass % of anisotropic ferrite is contained as magnetic powder. The shaft **30** has an outer diameter of $\Phi 6$ mm and an axial length of 330 mm. The material of the shaft **30** is SUM23 with nickel plating. For example, Aron Alpha (registered trademark) 800 series is used as the adhesive G. The amount of application of the adhesive G is 100 gm for example. The adhesive G is applied at a position along the center line C shown in the figure on the inner surface **12i** at the inside. The drying time of the adhesive G is at least 1 hour at room temperature.

[0069] Test results in the above-mentioned test conditions will be described with reference to FIGS. **8A** and **8B**. FIG. **8A** is a diagram showing an example of measurement results with bond strength and magnetic properties of the magnets **10** when the “groove width ratio” is at various levels. FIG. **8B** is a diagram showing an example of measurement results with bond strength and magnetic properties of the magnets **10** when the “groove depth ratio” is at various levels.

[0070] FIGS. **8A** and **8B** show measurement results of the magnetic properties (maximum magnetic force, magnetic center angle) of the first magnet **11d** and the bond strength between the first magnet **11d** and the shaft **30** when only a first magnet **11d** is bonded to the shaft **30**. Then, the measurement results of the magnetic properties of the second magnet **21** (see FIG. **3**) when the second magnet **21** is bonded to the shaft **30** in addition to the first magnet **11d** are shown. Note that the reference rotation angle (0°) in the “magnetic center angle” in FIGS. **8A** and **8B** is an angle overlapping the center line C of the first magnet **11d** in the circumferential direction D2 (see FIG. **6**).

[0071] According to FIG. **8A**, when the groove width ratio is 12% or more and 25% or less as described above, the magnetic properties of the first magnet **11d**, the bond strength, and the magnetic properties of the second magnet **21** are stable and are in the range of more favorable results. Therefore, if the groove width ratio of the magnet roller **1d** is in the range of 12% or more and 25% or less, the product performance is stably ensured, so an improvement in quality is achieved.

[0072] According to FIG. **8B**, when the groove depth ratio is 7% or more and 14% or less as described above, the magnetic properties of the first magnet **11d**, the bond strength, and the magnetic properties of the second magnet **21** are stable and are in the range of more favorable results. Therefore, if the groove width ratio of the magnet roller **1d** is in the range of 7% or more and 14% or less, the product performance is stably ensured, so an improvement in quality is achieved.

[0073] Note that the grooves **13** described in the first embodiment of the present disclosure are an example of the recess portions, and the shape of the recess portion is not limited to the groove shape. For example, the bottom of the recess portion may be curved, R-shaped, or rounded. Even if the recess portion is configured in this way, excess adhesive G of the applied adhesive G flows into the recess portion, and actions and effects similar to the above-mentioned actions and effects can be provided. Moreover, also in the second embodiment, as long as the groove width **13w** and the groove depth **13d** in the grooves **13** are in the above-mentioned range, the edges of the grooves **13**, the corners of the bottoms, or the like may be chamfered or rounded.

[0074] Note that in the above-mentioned first and second embodiments, the plurality of magnets **10**, **10d** is favorably molded from a blend material including magnetic powder and resin, and in addition, the magnetic powder may be configured to include ferrite. Accordingly, the magnets **10**, **10d** can be molded by injection molding or the like, so that the magnets **10**, **10d** can be molded with grooves **13** in the present disclosure.

[0075] Moreover, the plurality of magnets **10**, **10d** is favorably molded from a blend material including magnetic powder and resin. In addition, the magnetic powder may be constituted by a mixture of ferrite and rare-earth magnetic powder. In general, rare earth magnetic powders have stronger magnetic force and better temperature characteristics than ferrite magnetic powders. Therefore, if the magnetic powder is constituted by a mixture of ferrite and rare-earth magnetic

powder, it is possible to further enhance the compactness, lightness, and high performance in the operating environment, and to improve the degree of freedom in design.

[0076] Hereinabove, the embodiment of the present disclosure has been described with reference to the drawings. Note that the present disclosure is not limited to the above-mentioned embodiment, and may be carried out in various aspects without departing from the gist. For easy understanding, the drawings each schematically show configurations of elements mainly, and the thickness, the length, the number of items, the intervals, and the like of each component shown in the figure are different from the actual ones for the sake of convenience for creating the drawings. Moreover, the material, the shape, the dimensions, and the like of each component shown in the above-mentioned embodiment, and there are no particular limitations and various modifications can be made without substantially departing from the configurations of the present disclosure.

[0077] It should be understood by those skilled in the art that various modifications, combinations, sub-combinations and alterations may occur depending on design requirements and other factors insofar as they are within the scope of the appended claims or the equivalents thereof.

Claims

1. A magnet roller, comprising: a shaft; a plurality of magnets that is arranged in a circumferential direction of a shaft center of the shaft on an outer surface of the shaft; and an adhesive that bonds the shaft and the plurality of magnets to each other, wherein at least one of the plurality of magnets has a recess portion that is recessed with respect to a surface facing the outer surface of the shaft.
 2. The magnet roller according to claim 1, wherein the at least one of the plurality of magnets has a plurality of recess portions on the surface, and the plurality of recess portions are spaced apart from each other in the circumferential direction.
 3. The magnet roller according to claim 2, wherein a ratio of a sum of widths of the plurality of recess portions in the circumferential direction to a width of the surface in the circumferential direction is 12% or more and 25% or less.
 4. The magnet roller according to claim 2, wherein a ratio of a depth of each of the plurality of recess portions with respect to a dimension of the magnet in a radial direction of the shaft center is 7% or more and 14% or less.
 5. The magnet roller according to claim 1, wherein the plurality of magnets is molded from a blend material including magnetic powder and resin, and the magnetic powder includes ferrite.
 6. The magnet roller according to claim 1, wherein the plurality of magnets is molded from a blend material including magnetic powder and resin, and the magnetic powder is a mixture of ferrite and rare-earth magnetic powder.
 7. A developing roller, comprising: a magnet roller according to claim 1; and a tubular sleeve that houses the magnet roller and is rotatably supported around the circumferential direction.
 8. An image forming apparatus comprising a developing roller according to claim 7.
-